25/25 excellent work keep it up

Blessed Gebriel Michael Catholic Secondary School

2nd Semester Mathematics test one for Grade 10th

March, 2018 E.C Time Allowed 1:25'

Name: Bitanya Wondmagegnehu Sec: 10th B R.No. 12

I. Choose the best answer among the given alternatives.

C 1. If the terminal point of a given angle θ in standard postion is (-3,4) then which one is true about θ?

A. $\tan\theta = \frac{-3}{4}$ B. $\cot\theta = \frac{-4}{3}$ C. $sec\theta = \frac{-5}{3}$ D. *All*

C 2. Which one of the following is co- terminal with -56^0

A. -34^0 B. -124^0 C. 304^0 D. -304^0

AC 3. If θ *is an obtuse angl and* $cos\theta = \frac{-4}{5}$ then $\cot\theta =$ ____

A. $\frac{-3}{4}$ B. $\frac{3}{5}$ C. $\frac{-4}{3}$ D. $\frac{-4}{5}$

C 4. What is the trigonometric values of cot 810^0

A. 1 B. ∄ C. 0 D. -1

D 5. If $\cos\theta = \frac{8}{10}$ and θ *is* a 4th quadrant angle then what is the value of csecθ?

A. $\frac{-8}{6}$ B. $\frac{-8}{12}$ C. $\frac{-3}{5}$ D. $\frac{-5}{3}$

D 6. Which one of the following is true about any angle θ

A. $\sin^2\theta + cos^2\theta = 1$ B. $tan^2\theta + 1 = sec^2\theta$ C. $1 + cot^2\theta = csc^2$ D. *all* ~~exeetis~~

B 7. If $\cos\frac{\pi}{9} = sin\theta$ then what is the value of θ?

A. $\frac{9}{18}\pi$ B. $\frac{7}{18}\pi$ C. $\frac{6}{18}\pi$ D. $\frac{8}{18}\pi$

D 8. If $\sin\theta = cos\theta$ *and* θ is a 3rd quadrant angle then $\theta =$ _____

A. 235^0 B. 225^0 C. -135^0 D. B&C

D 9. Which one of the following is a quadrantal angle

A. -450^0 B. 540^0 C. 630^0 D. All of then

l=20m 10. What is the length of a ladder that leans against a building of height 10m in which the ladder makes 30^0 with the ground .

A. $8\sqrt{3}$ B. $24\sqrt{3}$ C. $16\sqrt{3}$ D. $16cm$

II. Give short answer only

11. The range of the cosine function is [-1, 1]

12. The domain of the sine function is All real no

13. The period of tangent function is π

14. The coterminal of -75^0 is -435^0

15. The reference angle of 290^0 is $\theta_r = 70^0$

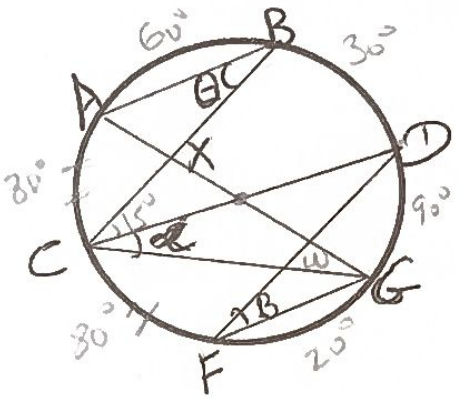
III. Work out (show all the necessary steps)

16. If θ *is an angle in stard position with terminal point* $p(-6,-8)$ *then find*

a. $\sin\theta$ $\quad$ *b.* $\cos\theta$ $\quad$ *C.* $\tan\theta$ $\quad$ *d.* $cse\theta$

17. A chord of a circle has length 8cm if the distance from the center to the mid point of the chord is 3cm then what is the length of the radius ?

18. Based on the figure below, Find the value of $\theta, \alpha, \beta, \omega$ & x where $\omega = A\hat{G}C$



. If AG is a diameter, arc AB=60^0, $\overset{\frown}{FG}$= 20, B$\hat{C}D$= 15 and arc $\overset{\frown}{AC}$= arc CF

(6, Answer

find r = Using Pythagores theorem $r^2 = x^2 + y^2 = r^2 = (-6)^2 + (-8)^2 = 36 + 64$

$\sqrt{r^2} = \sqrt{100}$ $\quad = r = 10$



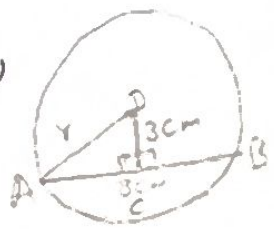
$\sin\theta = \frac{opp}{hypo} = \frac{-8}{-10}$

$\cos\theta = \frac{ads}{hypo} = \frac{-6}{10}$

$\tan\theta = \frac{opp}{ads} = \frac{-8}{-6} = \frac{8}{6}$

$csec\theta = \frac{1}{\sin\theta} = \frac{1}{-8/10} = \frac{-10}{8} = \frac{-5}{4}$

17,



Since 3cm is drawn to the center of the chord it bisect the chord and meets 90° then $\frac{1}{2}(AB) = \frac{1}{2}(8) = 4cm$

Using Pythagores theron $r^2 = 3^2 + 4^2 = r^2 = 9 + 16$ $\sqrt{r^2} = \sqrt{25}$

$r = 5cm$ (2)

18, AG is drameter so it creats a semi circle

arc $\overset{\frown}{FG} = 20°$ & $\overset{\frown}{AC} = \overset{\frown}{CF}$ & $\overset{\frown}{FG} + \overset{\frown}{AC} + \overset{\frown}{AC} = 180° = 20 + 2(\overset{\frown}{AC}) = 180° = \frac{2\overset{\frown}{AC}}{2} = \frac{160°}{2}$

$\overset{\frown}{AC} = 80°$ $\overset{\frown}{CF} = 80°$ so, $\theta = \frac{1}{2}\overset{\frown}{AC} = \frac{1}{2}(80°) = 40°$

$\omega = \frac{1}{2}\overset{\frown}{AC} = \frac{1}{2}(80°) = 40°$

$\overset{\frown}{AB} = 60°$ & $\angle BCD = 15° = \frac{1}{2}\overset{\frown}{BD}$ $= arc\ \overset{\frown}{BD} = 30°$ Since arc ABDG is semicircle

$\overset{\frown}{AB} + \overset{\frown}{BD} + \overset{\frown}{DG} = 180° = 60° + 30° + \overset{\frown}{DG} = 180°$, $\overset{\frown}{DG} = 90°$

$\alpha = \frac{1}{2}\overset{\frown}{DG} = \frac{1}{2}(90°) = 45°$

$\beta = \frac{1}{2}\overset{\frown}{DG} = \frac{1}{2}(90°) = 45°$

$\alpha = \frac{1}{2}(\overset{\frown}{BD} + \overset{\frown}{AC}) = \frac{1}{2}(30° + 80°) = \frac{1}{2}(110°) = 55°$